TABLE 4.1 Summary of Procedure to Determine the Far-Field Radiation Characteristics of an Antenna

1. Specify electric and/or magnetic current densities **J, M** [physical or equivalent (see Chapter 3, Figure 3.1)]
2. Determine vector potential components $A_\theta,\ A_\phi$ and/or $F_\theta,\ F_\phi$ using (3-46)–(3-54) in far field
3. Find far-zone **E** and **H** radiated fields ($E_\theta,\ E_\phi;\ H_\theta,\ H_\phi$) using (3-58a)–(3-58b)
4. Form either

a. $$\mathbf{W}_{\text{rad}}(r,\theta,\phi) = \mathbf{W}_{\text{av}}(r,\theta,\phi) = \frac{1}{2}\text{Re}[\mathbf{E}\times\mathbf{H}^*]$$
$$\simeq \frac{1}{2}\text{Re}\,[(\hat{a}_\theta E_\theta + \hat{a}_\phi E_\phi)\times(\hat{a}_\theta H_\theta^* + \hat{a}_\phi H_\phi^*)]$$
$$\mathbf{W}_{\text{rad}}(r,\theta,\phi) = \hat{a}_r\frac{1}{2}\left[\frac{|E_\theta|^2 + |E_\phi|^2}{\eta}\right] = \hat{a}_r\frac{1}{r^2}|f(\theta,\phi)|^2$$

or

b. $U(\theta,\phi) = r^2\ W_{\text{rad}}(r,\theta,\phi) = |f(\theta,\phi)|^2$

5. Determine either

a. $$P_{\text{rad}} = \int_0^{2\pi}\int_0^{\pi} W_{\text{rad}}(r,\theta,\phi) r^2 \sin\theta\, d\theta\, d\phi$$

or

b. $$P_{\text{rad}} = \int_0^{2\pi}\int_0^{\pi} U(\theta,\phi)\sin\theta\, d\theta\, d\phi$$

6. Find directivity using

$$D(\theta,\phi) = \frac{U(\theta,\phi)}{U_0} = \frac{4\pi U(\theta,\phi)}{P_{\text{rad}}}$$
$$D_0 = D_{\max} = D(\theta,\phi)|_{\max} = \frac{U(\theta,\phi)|_{\max}}{U_0} = \frac{4\pi U(\theta,\phi)|_{\max}}{P_{\text{rad}}}$$

7. Form *normalized* power amplitude pattern:

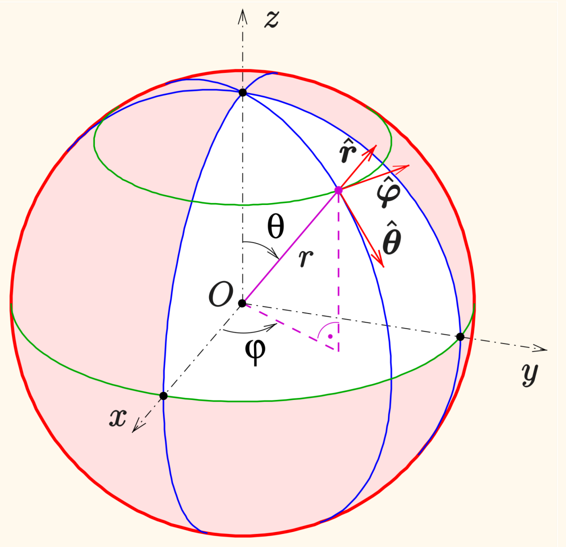
$$P_n(\theta,\phi) = \frac{U(\theta,\phi)}{U_{\max}}$$

8. Determine radiation and input resistance:

$$R_r = \frac{2P_{\text{rad}}}{|I_0|^2};\qquad R_{in} = \frac{R_r}{\sin^2\left(\dfrac{kl}{2}\right)}$$

9. Determine maximum effective area

$$A_{em} = \frac{\lambda^2}{4\pi}D_0$$



Electrical Length	L/λ where L is the largest dimension in antenna

Field Regions

Reactive Near Fields $\left(\frac{1}{(kr)^2}\ \&\ \frac{1}{(kr)^3}\right)$:

$r < r_1 = 0.16\lambda \quad (kr = 1)$

Radiative Near Fields $\left(\frac{1}{kr}\ \&\ \frac{1}{(kr)^2}\right)$: $r_1 < r < R_{ff}$

Radiative Far Fields $\left(\frac{1}{kr}\right)$: $r > R_{ff}$

EE4112 Definition:

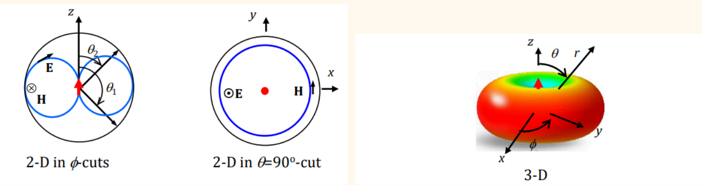
$$R_{ff} = \begin{cases} 2.5\lambda & for\ L < 2.5\lambda \\ \frac{2L^2}{\lambda} & otherwise \end{cases}$$

EE5308 Definition:

$$R_{ff} = \begin{cases} 3\lambda & \text{for } L \le 1.25\lambda\ (\mathrm{I}) \\ 2\frac{L^2}{\lambda} & \text{for } L \ge 1.25\lambda\ (\mathrm{II}) \end{cases}$$

Radiation Pattern

Electric Field Pattern



$$E(r, \theta, \phi) = [a_\theta F_\theta(\theta, \phi) + a_\phi F_\phi(\theta, \phi)]V\frac{e^{-jkr}}{r}$$

$$F_{\theta\ or\ \phi}(\theta, \phi) = \frac{|E_{\theta\ or\ \phi}(\theta, \phi)|}{|(E_{\theta\ or\ \phi})|_{max}}$$

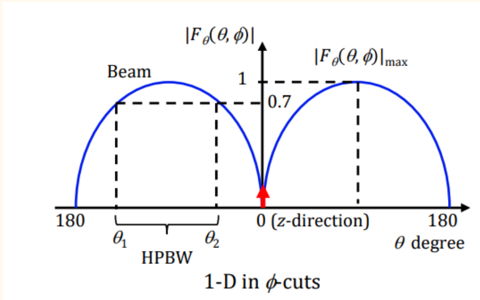
$$F_{\theta\ or\ \phi}(\theta, \phi)\{dB\} = 20log(F_{\theta\ or\ \phi}(\theta, \phi))$$

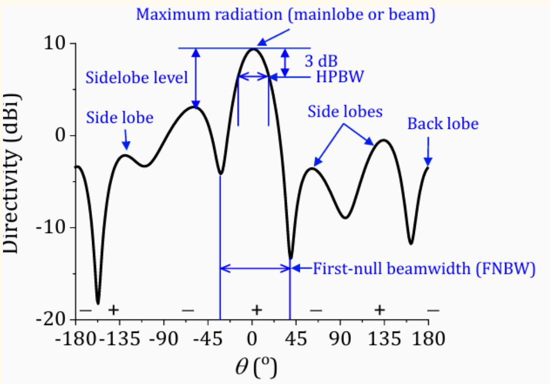
Power Pattern

$$P(\theta, \phi) = |F_{\theta\ or\ \phi}(\theta, \phi)|^2$$

$$P(\theta, \phi)\{dB\} = 10log(P(\theta, \phi)) = F_{\theta\ or\ \phi}(\theta, \phi)\{dB\}$$

Half Power Beamwidth (HPBW)





(a) Radiation pattern in a Cartesian coordinate system

Poynting Vector (farfield: r > R_{ff}) / Time Average Power Density Vector / Directional Energy Flux

$$S_{ave}(r,\ \theta,\ \phi) = \frac{1}{2} Re\{E(r,\ \theta,\ \phi) \times H(r,\ \theta,\ \phi)^{\square}\}$$

$$S_{ave}(r,\ \theta,\ \phi) = \frac{1}{2\eta} |E(r,\ \theta,\ \phi)|^2 a_k$$

$$S_{ave}(r,\ \theta,\ \phi) = \frac{\eta}{2} |H(r,\ \theta,\ \phi)|^2 a_k$$

Total Power Radiated

$$P_{rad} = Re\{P_{total}\} = Re\{\oint_s S_{ave} ds\}$$

$$\oint_s S_{ave} ds = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} S_{ave}(r,\ \theta,\ \phi)\ \ r^2 sin\theta\ d\phi\ d\theta$$

(Re is redundant for far field, not reactive component in free space)

Radiation Resistance

$$R_{rad} = \frac{2P_{rad}}{|I|^2} \qquad R_{rad} + R_{loss} = \frac{2P_{in}}{|I|^2}$$

Radiation Intensity

$$U(\theta,\ \phi) = r^2|S| = U_{max}|F(\theta,\ \phi)|^2$$

$$P_{rad} = \oint_{4\pi} U(\theta,\ \phi) d\square = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} U(\theta,\ \phi)\ \ sin\theta\ d\phi\ d\theta$$

Directivity

$$D(\theta,\ \phi) = \frac{U(\theta, \phi)}{U_{iso,\, ave}} = \frac{U(\theta, \phi)}{\frac{1}{4\pi} P_{rad}} = \frac{4\pi\, U(\theta, \phi)}{\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} U(\theta, \phi)\ \ sin\theta\ d\phi\ d\theta}$$

$$D_{max} = \frac{U_{max}}{U_{iso,\, ave}}$$

$$= \frac{U_{max}}{\frac{1}{4\pi} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} U_{max}|F(\theta, \phi)|^2\ \ sin\theta\ d\phi\ d\theta}$$

$$= \frac{4\pi}{\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} |F(\theta, \phi)|^2\ \ sin\theta\ d\phi\ d\theta}$$

$$D\{dBi\} = 10 log D_{max}$$

Estimation of Directivity for Pencil Beam (EE3104/5308)

Linear $D = \frac{32{,}400}{\Omega_b(\text{degree})^2} = \frac{32{,}400}{\text{HPBW}_1 \times \text{HPBW}_2}$

Estimation of Directivity for Pencil Beam (EE4101)

$$D = \frac{U_{\text{maximum}}}{U_{\text{average}}} \approx \frac{41,253}{\theta_{1r}^{3dB}\ \theta_{2r}^{3dB}}$$

(3dB beamwidth cited in degrees)

41253 is from $4\pi/\theta_1\theta_2$ with θ in radians

Derived for rectangular beam, better for narrowbeam

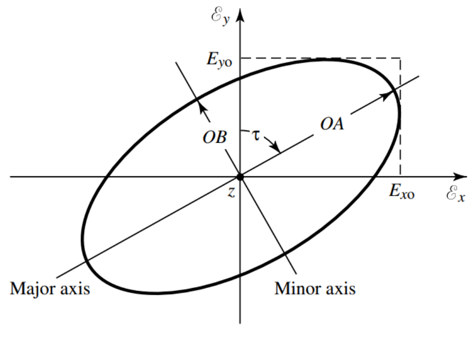
courses.egr.uh.edu/ECE/ECE5318/chapter2rev011611_STUDENT.pdf

Polarisation

A wave traveling in z-direction, its E-field in phasor at t and z:

$$\mathbf{a_E}E = \mathbf{a_x}E_x + \mathbf{a_y}E_y e^{j\Delta\phi}$$

- *Elliptical polarization*:
  - $\mathbf{a_E}E=\mathbf{a_x}E_x \pm j\mathbf{a_y}E_y$ as $\Delta\phi=\pm(1/2+n)\pi$, n=0, 1, 2…
  - $\mathbf{a_E}E=\mathbf{a_x}E_x+\mathbf{a_y}E_y e^{j\Delta\phi}$ as $\Delta\phi \neq \pm(1/2+n)\pi$, n=0, 1, 2…
- *Circular polarization*:
  $\mathbf{a_E}E = E_x(\mathbf{a_x} \pm j\mathbf{a_y})$ as $E_x=E_y$ and $\Delta\phi=\pm(1/2+n)\pi$, n=0, 1, 2…
- *Linear polarization*:
  $\mathbf{a_E}E=\mathbf{a_x}E_x+\mathbf{a_y}E_y$ for $\Delta\phi=\pm n\pi$, n=0, 1, 2…



(b) Polarization ellipse

Polarization	Orthogonal components	Phase shift, $\Delta\phi$	Amplitude ratio, $\lvert E_x\rvert/\lvert E_y\rvert$
elliptical	Both E_x and E_y	$\neq \pm n\pi$, n=0, 1, 2…	Any
		$\pm(1/2+n)\pi$, n=0, 1, 2…	Any but unity
circular	Both E_x and E_y	$\pm(1/2+n)\pi$, n=0, 1, 2…	Unity
linear	One or two of E_x and E_y	$\pm n\pi$, n=0, 1, 2…	Any

- Always plot and check thumb points to direction of propagation, if the rotation of E vector follows fingers then its RHCP or LHCP
- if propagating into +Z, $\Delta\phi > 0 \Rightarrow$ CCW
- if propagating into -Z, $\Delta\phi > 0 \Rightarrow$ CW

A plane wave propagating in z-direction:

$$\mathbf{E}=\mathbf{a_x}E_{x0}\cos(\omega t - kz)+\mathbf{a_y}E_{y0}\cos(\omega t - kz + \Delta\phi)$$

- *Axial Ratio (AR)*:

  The ratio of the major to minor axes of a polarization ellipse of an elliptically polarized wave.

  $$AR = \frac{\text{major axis }(OA)}{\text{minor axis }(OB)},\ AR_{\text{dB}} = \boxed{10}\log\left|\frac{E_{x0}}{E_{y0}}\right| \text{ dB}$$

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  - $1 \le AR \le \infty$ or $0 \text{ dB} \le AR \le \infty$ dB
  - $AR = 1$ for CP
  - $AR = \infty$ for LP
- $AR=\max\{AR\} \le 2$ or 3 dB for acceptable CP

For elliptical polarization, the curve traced at a given position as a function of time is, in general, a tilted ellipse, as shown in Figure 2.23(b). The ratio of the major axis to the minor axis is referred to as the axial ratio (AR), and it is equal to

$$\mathrm{AR} = \frac{\text{major axis}}{\text{minor axis}} = \frac{OA}{OB}, \quad 1 \le \mathrm{AR} \le \infty \tag{2-65}$$

where

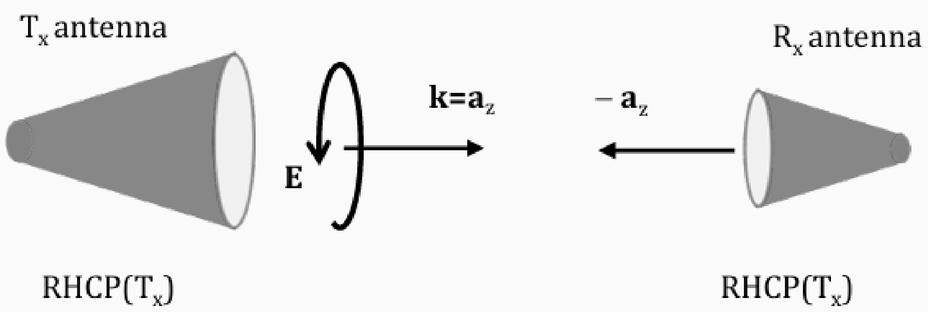
$$OA = \left[\tfrac{1}{2}\{E_{xo}^2 + E_{yo}^2 + [E_{xo}^4 + E_{yo}^4 + 2E_{xo}^2 E_{yo}^2 \cos(2\Delta\phi)]^{1/2}\}\right]^{1/2} \tag{2-66}$$

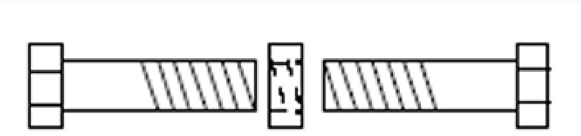
$$OB = \left[\tfrac{1}{2}\{E_{xo}^2 + E_{yo}^2 - [E_{xo}^4 + E_{yo}^4 + 2E_{xo}^2 E_{yo}^2 \cos(2\Delta\phi)]^{1/2}\}\right]^{1/2} \tag{2-67}$$

The tilt of the ellipse, *relative to the y axis*, is represented by the angle τ given by

$$\tau = \frac{\pi}{2} - \frac{1}{2}\tan^{-1}\left[\frac{2E_{xo}E_{yo}}{E_{xo}^2 - E_{yo}^2}\cos(\Delta\phi)\right] \tag{2-68}$$

When the ellipse is aligned with the principal axes [$\tau = n\pi/2, n = 0, 1, 2, \ldots$], the major (minor) axis is equal to $E_{xo}(E_{yo})$ or $E_{yo}(E_{xo})$ and the axial ratio is equal to E_{xo}/E_{yo} or E_{yo}/E_{xo}.





RHCP Tx antenna (definition based on Tx direction):

- Incident wave: $\mathbf{a}_i = (\mathbf{a}_x - j\mathbf{a}_y)/\sqrt{2}$
- Receive antenna: $\mathbf{a}_r = (\mathbf{a}_x + j\mathbf{a}_y)/\sqrt{2}$
- Polarization matching efficiency: $e_{pm} = |\mathbf{a}_i \cdot \mathbf{a}_r|^2 = \left|\frac{(\mathbf{a}_x - j\mathbf{a}_y)(\mathbf{a}_x + j\mathbf{a}_y)}{2}\right|^2 = 1$

$\therefore$ perfect polarization matching

Bandwidth (S11 < -10dB or VSWR < 2:1)

$Absolute\ BW\ = f_h - f_l$	$Fractional\ BW\ = \frac{f_h - f_l}{\frac{f_h + f_l}{2}} \times 100\%$ $Fractional\ BW\ = \frac{BW}{f_{center}} \times 100\%$ for BW < 100%
When scaling an antenna in dimension and assuming the $f_h\, f_l$ scale linearly, the absolute BW changes, but fractional and ratio BW stays the same	

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